

## REQUIREMENTS ENGINEERING

# LECTURE 2: VIEWS & PRODUCT VISION



Giảng viên biên soạn: Th.S Nguyễn Đức Quỳnh Mi



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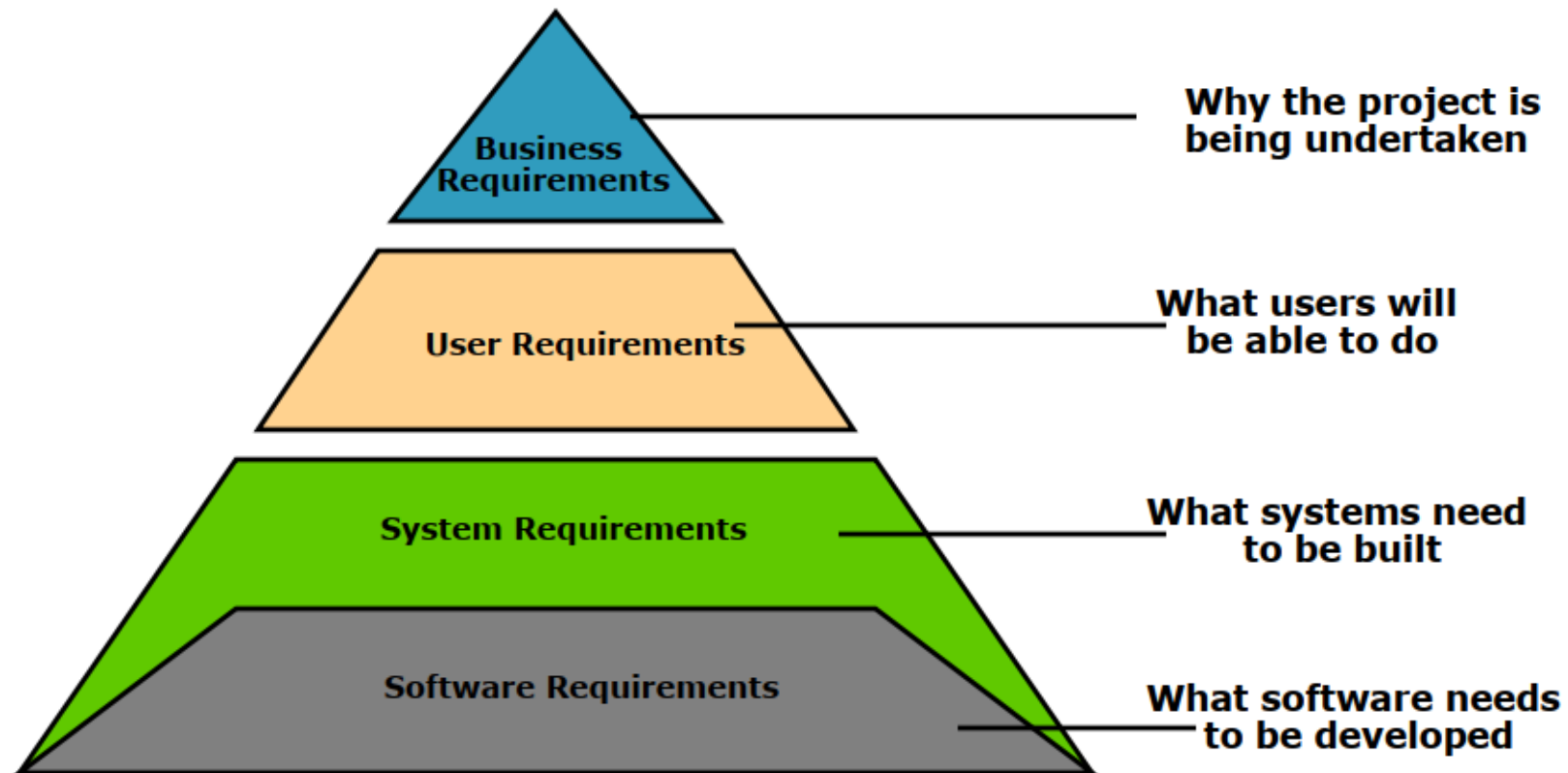
# What Are Requirements?\*

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- A condition or capability needed by a user or customer to solve a problem or achieve an objective.
- A condition or capability that must be met by a system or system component to satisfy a contract, standard, specification, or other formally imposed document.
- A document representation of a condition or capability.

*\* Based on IEEE Standard Terminology*

# Where Do Requirements Come From?





# **Business Requirements**

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- Statements of the business rationale for authorizing the project. They include a vision for the software product to be built that is driven by business goals, objectives, and strategy.
- Business requirements describe the high-level purposes and needs that the software product will satisfy, the view of its accomplishments for the users, its features, functions, and capabilities from the **Business point of view**.
- Business requirements are documented in the Project Charter, Vision or Scope of the project. Sometimes it can be found in the Statement of Work, Memo of Understanding (MoU) or Request for Proposal (RFP).



# Users Requirements

- The definition of the entire system (hardware and software) from the **User's point of view**. They describe the task that users need to do with the system to accomplish a business function.
- User requirements are the bridge between the business goals (Business language) and the system requirements (Technical language).
- Software engineers must understand how users will use the system and derive their requirements from the user requirements document.
- User requirements can be found in User Requirements Document (URD), Concept of Operations (Con OP), User Scenarios / Use-cases or Product

Backlog document.

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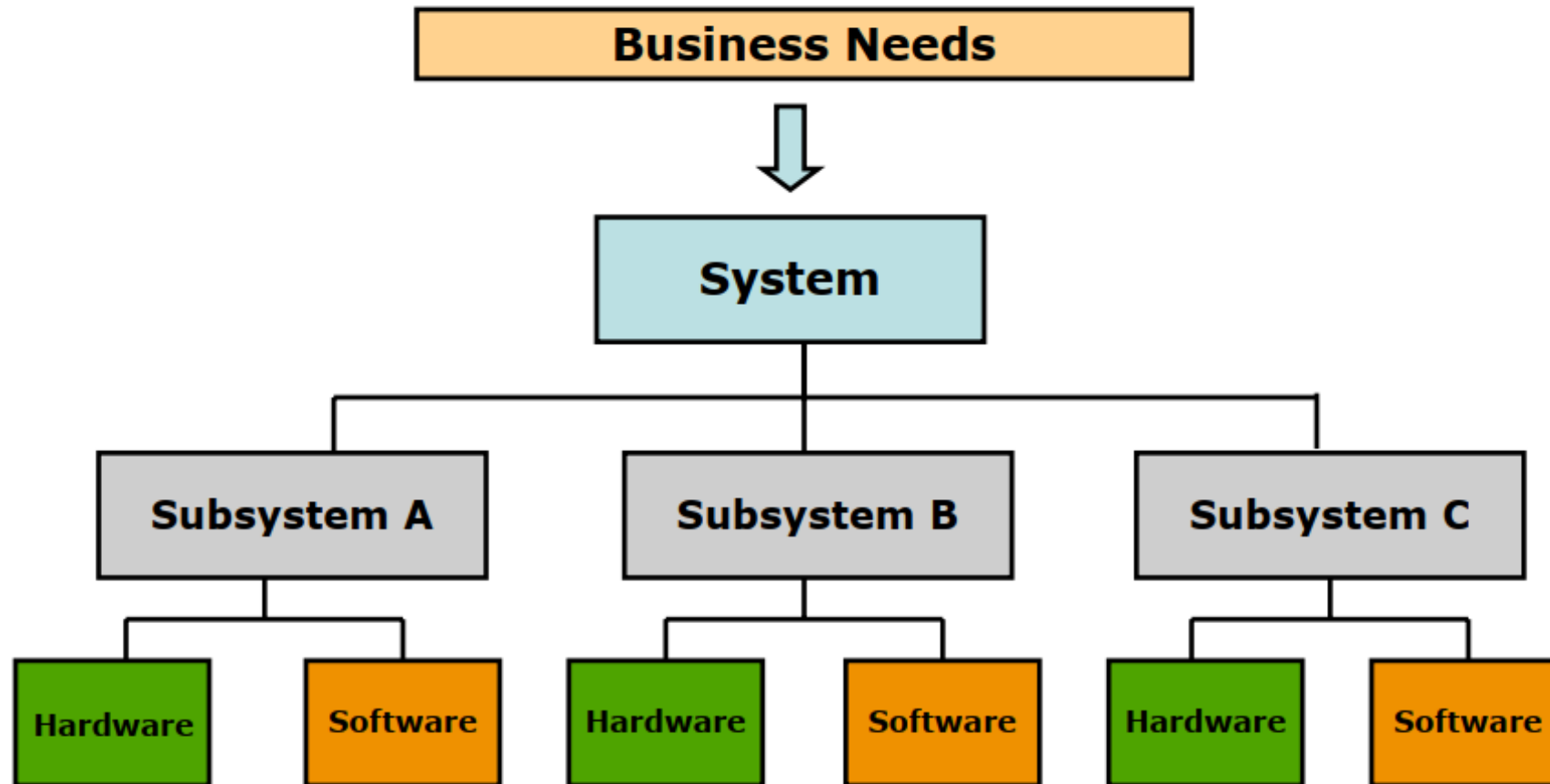


# System Requirements

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- Detailed descriptions of all functional as well as non-functional requirements that the systems (Hardware and Software) must do to meet business and user needs.
- System requirements define the top-level requirements for allocation to hardware, software or subsystems from the System Engineers' point of view.  
(Manager, Architect, Designer)
- System requirements serve as a communications channel to users, procurement organizations, as well as to system architects who are concerned with the development of system elements or components.

# System View





# Software Requirements

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- Detailed descriptions of all functional as well as non-functional requirements that the software must do to meet system requirements and user needs from the **Developer's point of view**.
- Software requirements establish an agreement between technical people and business people on what the product or application must do while staying within the constraints of system architecture and hardware limits.
- Software Requirements are documented in the Software Requirements Document (SRS), Detailed Requirements Specification or Functional Specification.





# Functional vs. Non-Functional

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- Software systems are characterized both by their functional behavior (what the system does) and by their nonfunctional behavior (how the system behaves with respect to some observable attributes such as reliability, reusability, maintainability, etc.)
- **Functional requirements** describe **what the system/software must do**. Function is “**What**” a system does, Not “**How well**” it does it.
- **Non-functional requirements (Quality Attributes)** are **global constraints** on a system such as development costs, operational costs, performance, reliability, maintainability, portability, robustness, etc...
  - *Often known as the “ilities”.*
  - *Usually, cannot be implemented in a single module of a program*



# Constraints

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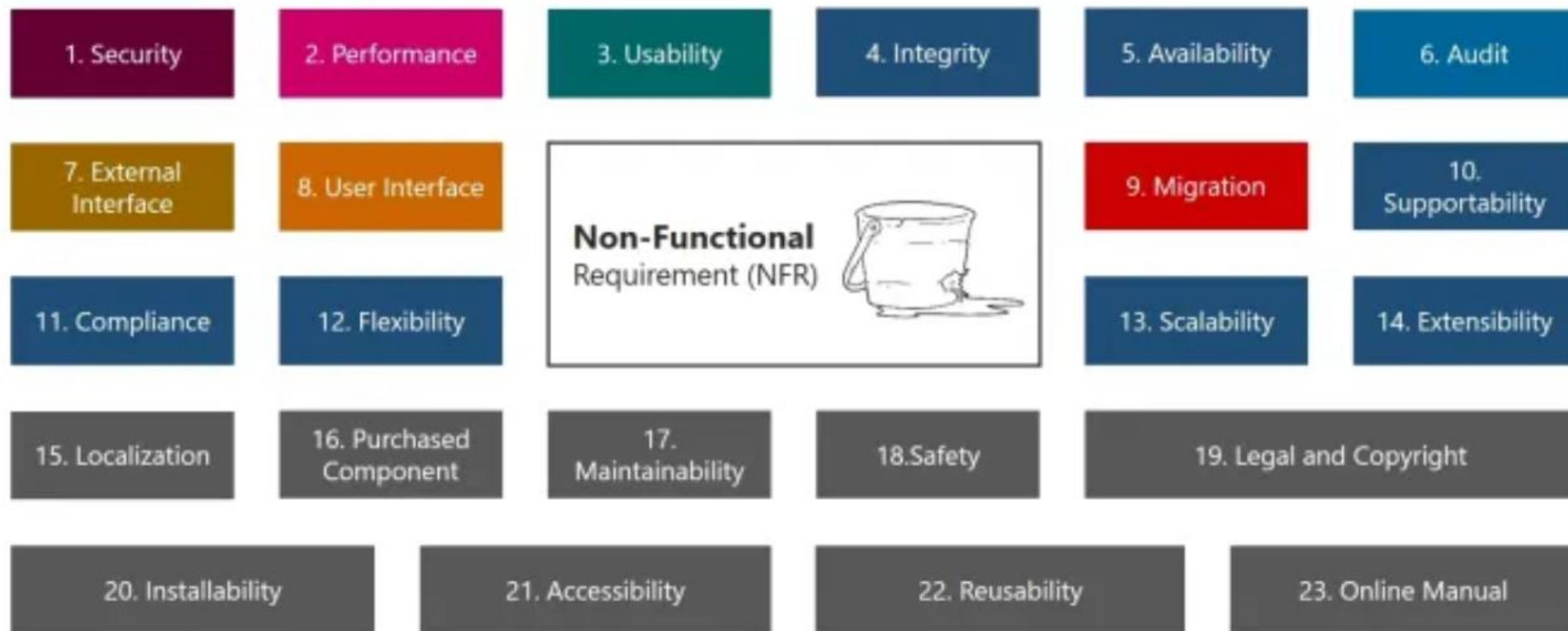
A constraint is a requirement specification that explicitly and intentionally restricts the engineering process, systems operation, or its lifecycle

There are two types of constraints:

- **Binary Constraints** are statements like ‘Must and Must not’. These are called Demands (must) and Vetoes (must not).
- **Scalar Constraints** are specified using a scale of measure, to limit some resources or capability in some way.

*Ex: The system must respond within 20 seconds after logon.*

# Quality Attributes(Non-Functional Requirements )



Thinhnotes.com

Source: <https://thinhnotes.com/chuyen-nghe-ba/23-loai-non-functional-requirement-troi-noi-it-ai-de-y/>



# Examples – Quality Attributes

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## Usability – Khả năng “Dễ sử dụng”

- *User muốn xem hàng tồn kho thì thay vì user chọn menu On-hand >> rồi chọn kho để xem, thì hãy hiển thị số lượng tồn kho ngay bên cạnh tên kho để user có thể xem được ngay, mà không cần nhấp thêm lần nào nữa.*
- *Phần mềm phải giảm thiểu tối đa thao tác của users trên màn hình để họ hoàn thành 1 task trên hệ thống nhanh nhất có thể. Ví dụ từ 5 chạm, giảm xuống còn 3 chạm là có thể order xong ly trà sữa. Hoặc nút bấm có label rõ ràng để user hiểu ngay được hành động sau khi bấm nút.*
- *Kiểm tra kiểu kiểu dữ liệu ngay trên từng field, và cảnh báo ngay nếu có lỗi, đừng bắt user nhập đi nhập lại nhiều lần.*

Source: <https://thinhnotes.com/chuyen-nghe-ba/23-loai-non-functional-requirement-troi-noi-it-ai-de-y/>



# Examples – Quality Attributes

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**Performance – Hiệu suất hoạt động của hệ thống, thường được đo lường bởi các tiêu chí:**

- Thời gian phản hồi cho một transaction
- Số lượng transaction thực hiện được trong mỗi giây
- Công suất (capacity), ví dụ số lượng transaction mà hệ thống có thể lưu trữ/ thực hiện cho mỗi đối tượng.
- Hoặc các yếu tố về tài nguyên sử dụng như: RAM, dung lượng DB...

*Ví dụ: Tất cả những màn hình input và output dữ liệu cần phải được sẵn sàng để hiển thị cho người dùng **trong vòng 3 giây**, với điều kiện tải và connection giữa client/server là bình thường, cụ thể:*

- **Điều kiện tải bình thường:** 30 CCU (concurrent user – người dùng đồng thời) khi không dùng cân bằng tải.
- **Điều kiện server tối thiểu:** Intel Core i5, 4GB RAM, 500GB hard disk.
- **Client/ Server Connection:** 500KB/s

Source: <https://thinhnotes.com/chuyen-nghe-ba/23-loai-non-functional-requirement-troi-noi-it-ai-de-y/>

# Functional vs. Non-Functional

## Example: Ứng dụng Ví điện tử Momo



### Chuyển - nhận tiền

Miễn phí, bảo mật, chuyển trong 1s



### Nạp tiền điện thoại

Top up - Mã thẻ cào - Data 3G/4G



### Du lịch - Đi lại

Vé máy bay - Tàu/Xe - Đặt khách sạn



### Sống tốt cùng MoMo

Trái Tim MoMo - Heo Đất MoMo



### Thanh toán hóa đơn

Điện - Nước - Internet - Chung cư



### Game / Ứng dụng

Thẻ Game - Google Play - App Store



### Vé xem phim

Đặt mua vé xem phim



### Vé số

Vietlott - Điện Toán - XS Kiến Thiết



### Thanh toán tại quầy

Siêu thị - Mua sắm - Xăng dầu



### Tài chính / Bảo hiểm

Túi Thần Tài - Ví Trả Sau - Bảo hiểm



### Mua sắm trực tuyến

Thương mại điện tử - Đi chợ online



# Understand Business Requirements

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- By understanding the "business requirements" you can build a "vision and scope" document that describes the background, leading to the decision to develop a new or modified system and a description of the system to be developed.
- You must consider having iterative scoping meetings with stakeholders to generate more detailed and creative thinking about the problem that in turn can affect the scope.
- Understanding **business problems** is the first step to capture better requirements. If you do not know the business problems that you are going to solve, you will never succeed.
- You must be able to describe the proposed system or situation and define the problem that you are going to solve.
  - *What is the proposed system used for?*
  - *The purpose of improving (in case of maintenance or adding more functions) or replacing the current system?*

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# Understand Business Requirements

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- You must review all the requirements that you captured from stakeholders and align them with business goals and objectives.
- You must analyze all requirements and document them, then review with stakeholders to make sure that you have captured their needs correctly.
- You must resolve **conflicts** among requirements between different stakeholders as early as possible.



# Requirements Preparation ...

Preliminary & Set the Stage:





# Define Product Vision

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- A vision statement is a **concise statement** that defines the **what, why, and who** will **buy or use** the final product from the business point of view.
- The vision statement will help you to:
  - Ensure that the product definition aligns with the business goals and objectives.
  - Identify stakeholders – Who will buy or use the product?
  - What the product will do for its stakeholders?
  - What are the reasons to buy or use this product?
  - How can this product be distinguished from others?
  - Describe the state of the business – its strengths, weaknesses.
  - Describe the opportunities and risks to the business.



# Define Product Vision

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- Insert statement to <>
- ***For*** <target customer> ***Who*** <statement of need> ***the*** <product name> ***is a*** <product category> ***that*** <benefit statement> ***unlike*** <competitive analysis>, ***our product or service*** <product differentiation>.
- Review and check if the vision aligns with the business goals and objectives.



# Example 1 – Product Vision

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- **For** software users **who** need to request information from databases, **the** ABC Data Management **is** an information system that provided a single point of access to all data and information stored anywhere in company's multiple databases. **Unlike** other systems, **our system** not only locates, selects and provides information, but also generates required reports when needed, for use for further investigation and documentation.



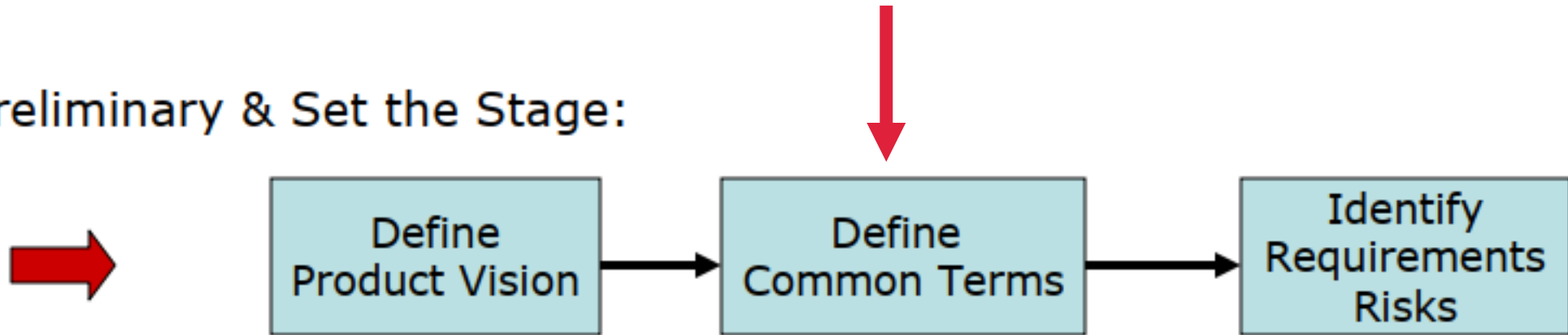
## Example 2 – Product Vision

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MoMo là ứng dụng Ví điện tử trên điện thoại thông minh của Công ty Cổ phần Dịch vụ Di động Trực tuyến (M\_Service), có mặt trên 2 hệ điều hành iOS và Android. Là nền tảng thanh toán di động, Ví MoMo thúc đẩy nền kinh tế không tiền mặt và cung cấp cho khách hàng trải nghiệm thanh toán một chạm (One Touch Payment) với hơn hàng trăm tiện ích dịch vụ, bao gồm Chuyển tiền, Thanh toán hóa đơn, Mua vé máy bay, Mua vé xe lửa, Vé xem phim, Thu-Chi hộ và Thương mại trên di động.

# Requirements Preparation ...

Preliminary & Set the Stage:





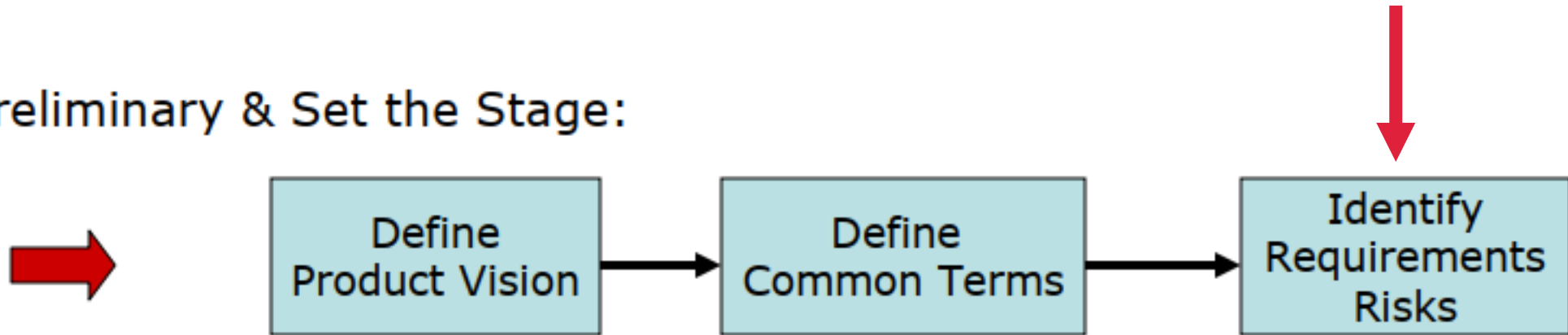
# Define Common Terms

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- Since stakeholders and software teams may not understand each other because each is using different terms, or use the same term with different meanings; it is important to build a dictionary of common terms relevant to the product being built to improve communication during requirements gathering.
- Improve a shared understanding of the problem space.
- Enable business people to inform technical people about important business concepts.
- Improve communication by eliminating misunderstanding in what business concepts mean.

# Requirements Preparation ...

Preliminary & Set the Stage:







# Requirements Risks

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- Lack of user involvement
- Unclear, ambiguous requirements
- Conflicting requirements
- Missing requirements
- Unrealistic expectations
- Constantly changing requirements
- New technology, techniques and tools
- New business model, new market



# Mitigate Requirements Risks

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- Requirements risk mitigation is a process to assess requirements-related risks and identifies actions to avoid or minimize those risks.
- Requirements risk mitigation also helps strengthen project team and stakeholder communication and helps the team prepare for or prevent obstacles to successful requirements development.
- Because requirements are critical to the project, identifying and solving these risks can have high impact on the project success



# Summary

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- Types of requirement: **Business Requirements, Users Requirements, System Requirements and Software Requirements**
- Functional Requirements vs Quality Attributes
- Before gathering requirements, you must define the product vision, common terms and mitigate risks.

*Thank you*  
*It's time for your questions*

